



Grey Relational Analysis of Carbon Emission Drivers in the Yellow River Basin

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Abstract: The achievement of carbon peaking and carbon neutrality has been identified as a central strategic priority for sustainable development in China. As a key national energy base and ecological barrier, the Yellow River basin has been subjected to substantial pressure for carbon emission mitigation. To elucidate the underlying driving mechanisms of carbon emissions, a panel data-based grey relational analysis framework was constructed using data from nine provinces (autonomous regions) within the basin over the period 2010–2021. The correlation intensity between carbon emissions and five principal influencing factors—regional gross domestic product, energy mix, added value of the secondary industry, year-end permanent resident population, and urban population—was quantified from temporal, cross-sectional, and integrated perspectives. The results indicate that the comprehensive grey relational degree follows the order: energy mix (0.90) > year-end permanent resident population (0.86) > urban population (0.85) > added value of the secondary industry (0.82) > regional gross domestic product (0.77), with energy mix identified as the dominant driver of carbon emissions. From a temporal perspective, the correlation between regional gross domestic product and carbon emissions was observed to decline steadily from 0.87 in 2010 to 0.68 in 2021, suggesting a progressive decoupling of economic growth from carbon emissions. In contrast, the correlation associated with energy mix remained consistently above 0.88, indicating limited advancement in energy structure transformation. Cross-sectional analysis reveals pronounced regional heterogeneity, with emission-driving patterns categorized into three types: energy–industry dominated, population–urbanization driven, and multi-factor integrated. These findings provide robust empirical evidence for the formulation of differentiated carbon mitigation strategies and offer critical insights for advancing ecological protection and high-quality development within the Yellow River basin.

Keywords: Panel data; Grey relational analysis; Yellow River basin; Carbon emissions; Low-carbon development

1. Introduction

In recent years, global warming has become a key topic in international academic research. The continuous accumulation of greenhouse gases is the primary cause of climate warming, and carbon emissions play a core role in this process. To cope with multiple crises caused by rising temperatures, countries across the globe have taken active measures. China has formulated differentiated governance policies tailored to local conditions and implemented them step by step. With overall coordinated planning, China has strengthened ecological protection, advanced green economic transformation and improved people's livelihood. It aims to build a new pattern of ecological conservation and sustainable development in the Yellow River basin. Therefore, it is of great significance to study the influencing factors of carbon emissions in the basin. It is expected that the research can improve the theoretical basis for the formulation of energy conservation and emission reduction policies. It also

provides practical references for governments to formulate targeted low-carbon strategies in line with regional realities.

1.1 Research on Carbon Emissions

Carbon emissions are critical to global sustainable development and humanity's shared future, and have attracted extensive scholarly attention across diverse disciplines. Researchers have focused on carbon emission projection and driver identification to support the formulation of energy-conservation and emission-reduction policies worldwide.

Zhang & Xu (2025) developed a carbon emission simulation framework for evaluating mobility-as-a-service projects, and built a prediction model by integrating the mobile-source emission framework of the Intergovernmental Panel on Climate Change with a multi-agent system. Rao et al. (2025) proposed a new carbon emission prediction method that combines three urban landscape indices with conventional socioeconomic variables. Using improved Stochastic Impacts by Regression on Population, Affluence, and Technology (STIRPAT) and Patch-generating Land Use Simulation models, they projected future socioeconomic and urban form changes, and estimated carbon emissions for 21 cities in Guangdong Province from 2025 to 2060. Yu et al. (2025) applied a backpropagation neural network, emission factor coefficients, the Tapio decoupling framework, and the logarithmic mean Divisia index model to analyze carbon emissions from crop planting and farmland soil in Heilongjiang Province from 2003 to 2022.

Ma et al. (2025) used panel data from 2005 to 2022 to explore the decoupling relationship between industrial carbon emissions and economic growth in the Beibu Gulf Urban Agglomeration, and identified driving factors using the Tapio decoupling model and the logarithmic mean Divisia index decomposition. The findings indicate that integrating parks and commercial facilities with other urban amenities significantly reduces carbon emissions in both sectors. This mitigation effect operates through two mechanisms: functional complementarity and environmental enhancement.

Previous studies have provided valuable references for in-depth carbon emission research in terms of research perspectives and methodological choices. Although relevant research in China started relatively late, it has developed rapidly and established a sound theoretical and empirical system. Nevertheless, most existing studies focus on economically developed regions such as the Yangtze River Delta and the Pearl River Delta. Research on the Yellow River basin remains limited by poor data accessibility and insufficient analysis of regional heterogeneity. In addition, traditional econometric methods require large sample sizes, making it difficult to effectively address missing statistical data in some provincial regions of the Yellow River basin.

1.2 Research Progress on Panel Data Grey Correlation Analysis

Grey system theory specializes in solving uncertain problems with small samples and poor information. As a core method of this theory, grey relational analysis evaluates the correlation degree between various factors by measuring the geometric similarity of data sequences, and it has been widely adopted in socioeconomic research. With the popularization of panel data, scholars have further extended grey relational analysis from traditional two-dimensional time series to three-dimensional panel data research. Zhang & Liu (2010) integrated surface cluster analysis (used to characterize the geometric features of panel data) with the principles of grey correlation analysis. They examined the geometric similarity of multiple indicators within a three-dimensional panel data framework, and proposed the extended grey correlation degree based on matrix construction. Qian et al. (2013) extended the calculation of grey correlation degree from vector space to matrix space, and developed a grey matrix correlation analysis model that captures the spatio-temporal characteristics of multi-index panel data. Wu & Liu (2013) expanded the convex correlation degree from a two-dimensional to a three-dimensional framework. Based on the positive semi-definiteness of the Hessian matrix, they defined three-dimensional convexity and put forward the three-dimensional grey convex correlation degree. Liu et al. (2014) used the grid method to describe the geometric characteristics of three-dimensional panel data, and established a grey grid correlation model. Cui & Liu (2015) evaluated the similarity of development rate indices between factor matrices and system characteristic behavior matrices from spatio-temporal perspectives. They then proposed a grey matrix-similar correlation model for panel data.

Grey relational analysis has some recent research applications. Li et al. (2025b) combined the multi-attribute border approximation area comparison method with grey relational analysis to develop an improved failure mode and effects analysis model. This model significantly improves the accuracy and reliability of risk assessment for digital transformation projects in the construction industry. Xiong et al. (2021) investigated the driving factors of carbon emissions in seven provinces and cities in East China. They built a panel data-based grey matrix similar correlation model and calculated the time-series and cross-sectional grey correlation degrees of various factors from 2005 to 2016. Candan & Toklu (2022) proposed an evaluation approach for sustainable industrialization performance in European Union countries, which integrates the spherical fuzzy analytic hierarchy process and

grey relational analysis. In addition, Rajesh (2022) developed a new advanced grey relational analysis method by combining grey theory with event analysis, which was applied to measure the resilience level of supply chains.

1.3 Research Content and Methods

Taking the panel data of nine provinces and autonomous regions in the Yellow River basin from 2010 to 2021 as the research sample, this study selects core indicators such as economic development, industrial structure, energy structure and population size. A panel grey correlation model is established to measure the correlation between carbon emissions and their influencing factors from temporal, cross-sectional and comprehensive dimensions. By identifying provincial differential characteristics, this study clarifies the driving mechanisms of carbon emissions in the Yellow River basin, providing references for regions to formulate differentiated energy conservation and emission reduction policies.

2. Construction of Grey Correlation Model of Panel Data

2.1 Selection of Carbon Emission Influencing Factors

A great number of researchers have delved into the realm of carbon emissions. Ehrlich & Holdren (1971) set off with a fundamental method by envisaging the environmental influence of human social and economic undertakings, and they put forward the Impact = Population $\times$ Affluence $\times$ Technology model, which associated carbon emissions with population quantity, affluence and technological progress. Dietz & Rosa (1994) built on this model and worked out the STIRPAT model, a stochastic version that regarded environmental impact as a function of population, wealth and technology within a regression setup. Wang et al. (2025) made use of the STIRPAT model to look into the embodied carbon emissions of residential buildings at the materialization stage and took into account factors like population quantity, income level, industrial structure and energy intensity. Chen et al. (2025) employed an improved STIRPAT and genetic algorithm–bidirectional long short-term memory model to predict the time when carbon emissions would reach their peak across China's provinces. It was found out that permanent population, per capita gross domestic product, energy intensity, secondary industry gross domestic product index, coal consumption proportion and urbanization rate were crucial factors. The results indicate that urbanization and industrial upgrading are the main drivers of carbon emissions, while population density is the most significant inhibiting factor.

Liu et al. (2025) used the logarithmic mean Divisia index model to analyze China's carbon emission elements and mitigation approaches and covered aspects such as population quantity, economic development, industrial structure, energy intensity, energy structure and carbon emission coefficient. Li et al. (2025a), in their study on the synergistic effect of reducing air pollutants and CO₂ emissions in China and its influencing factors, found that total energy consumption, energy consumption intensity, employment in the secondary industry, and the share of the secondary industry's added value in regional gross domestic product all inhibit the synergistic reduction of air pollutants and CO₂. He et al. (2025), taking the 28 member states of the European Union as a case study to examine the influencing factors of carbon emissions from final consumption, found that economic output, carbon structure, and population change are significant emission factors. And overall, this current study identified carbon emissions (million tons), gross domestic product (Chinese Yuan (CNY) 100 million), energy structure (%), added value of the secondary industry (CNY 100 million), total population (ten thousand) and urban population (ten thousand) as the main determinants of carbon emissions.

2.2 Data Sources

The current study centered on the Yellow River basin geographically and put stress on quantifying carbon emissions and identifying crucial factors across nine provinces: Shanxi, Inner Mongolia, Shandong, Henan, Sichuan, Shaanxi, Gansu, Qinghai, and Ningxia. Carbon emission figures came from the China carbon accounting databases and more empirical data were collected from the statistical yearbooks released by the local statistical agencies of these provinces between 2010 and 2021, which made the research results strong and reliable. In addition, due to the lack of available data, the energy structure for Shanxi Province was calculated using a framework and model established by several researchers (Geng et al., 2004; Guan et al., 2006; Wang et al., 2011; Zhong et al., 2007), thus filling the void in direct statistical data.

2.3 Construction of the Panel Data Grey Correlation Model

In this study, the examination of carbon emissions among the nine provinces and autonomous regions adjacent to the Yellow River basin was carried out and conventional analytical techniques like correlation and regression analysis were frequently used in such investigations. But they required a large sample size and often had difficulty

dealing with the intricacies of panel data which included both time-related and space-related aspects. Therefore, this research focused on the time-space combination within the panel data framework and was conducted using the method put forward by Xiong et al. (2021), a grey correlation analysis designed for panel data, to identify the factors affecting carbon emissions in the nine areas near the Yellow River basin.

2.3.1 Matrix representation of panel data

Definition 1: Consider a dataset made up of S subjects, each of which is described by N indicators and has observations noted down over a span of T time intervals. Let the observed value of the j -th indicator for the i -th subject at the t -th time point be denoted by $x_i(s, t)$, then the data matrix for the indicators could be shown as X with every element matching the relevant observed value.

$$X_i = \begin{bmatrix} x_i(1,1) & x_i(1,2) & \cdots & x_i(1,T) \\ x_i(2,1) & x_i(2,2) & \cdots & x_i(2,T) \\ \vdots & \vdots & \ddots & \vdots \\ x_i(S,1) & x_i(S,2) & \cdots & x_i(S,T) \end{bmatrix} \quad (1)$$

where, $i = 1, 2, \dots, N$; $S = 1, 2, \dots, S$; and $T = 1, 2, \dots, T$. The panel data can be expressed as follows:

$$X = \{x_1(s, t), x_2(s, t), \dots, x_n(s, t)\} \quad (2)$$

When the scalar S is equal to one, it indicates that there is only one research unit and the panel data includes a time series of different indicators related to this unit. On the contrary, when the scalar T is equal to one, it means there is just one point in time and the panel data show the cross-sectional values of various indicators for all subjects at that particular time.

2.3.2 Dimensionless processing of panel data

Because indicators have different dimensions and orders of magnitude, dimensionless processing is needed to eliminate their influence on the analysis results. Common dimensionless methods include the initial value method and the mean value method. All indicators in this study show clear time trends. Therefore, the initial value method is adopted. The initialization operator can be specified as follows:

$$d_i = \frac{1}{x_i(1,1)} \quad (3)$$

Initialization of the data matrix leads to the following:

$$d_i x_i = \begin{bmatrix} 1 & d_i x_i(1,2) & \cdots & d_i x_i(1,T) \\ d_i x_i(2,1) & d_i x_i(2,2) & \cdots & d_i x_i(2,T) \\ \vdots & \vdots & \ddots & \vdots \\ d_i x_i(S,1) & d_i x_i(S,2) & \cdots & d_i x_i(S,T) \end{bmatrix} \quad (4)$$

In the above matrix, $i = 1, 2, \dots, N$; $S = 1, 2, \dots, S$; and $T = 1, 2, \dots, T$.

Table 1 and Table 2 show the dimensionless processing results of the panel data.

Table 1. Initialized panel data of carbon emissions

Year	Shanxi	Inner Mongolia	Shandong	Henan	Sichuan	Shaanxi	Gansu	Qinghai	Ningxia
2010	1	1.098	1.741	1.188	0.614	0.521	0.295	0.074	0.217
2011	1.071	1.382	1.82	1.193	0.616	0.529	0.306	0.078	0.314
2012	1.16	1.469	2.019	1.224	0.782	0.621	0.357	0.104	0.316
2013	1.177	1.366	1.837	1.14	0.814	0.632	0.372	0.111	0.333
2014	1.153	1.389	1.894	1.26	0.813	0.66	0.381	0.113	0.334
2015	1.068	1.391	1.976	1.22	0.769	0.657	0.37	0.119	0.328
2016	1.098	1.402	1.997	1.206	0.738	0.63	0.356	0.131	0.323
2017	1.176	1.519	1.933	1.161	0.737	0.634	0.351	0.124	0.413
2018	1.253	1.674	2.085	1.135	0.685	0.639	0.377	0.12	0.443
2019	1.306	1.837	2.167	1.065	0.729	0.685	0.38	0.12	0.491
2020	1.349	1.942	2.153	1.096	0.711	0.717	0.407	0.111	0.523
2021	1.42	1.951	2.191	1.119	0.728	0.784	0.438	0.13	0.544

Table 2. Initialized panel data of provincial gross domestic product

Year	Shanxi	Inner Mongolia	Shandong	Henan	Sichuan	Shaanxi	Gansu	Qinghai	Ningxia
2010	1	0.921	3.81	2.544	1.935	1.106	0.443	0.129	0.177
2011	1.224	1.062	4.387	2.956	2.364	1.367	0.541	0.154	0.217
2012	1.312	1.176	4.825	3.253	2.687	1.588	0.606	0.172	0.239
2013	1.346	1.279	5.317	3.553	2.978	1.786	0.675	0.192	0.261
2014	1.358	1.365	5.703	3.883	3.245	1.954	0.732	0.208	0.278
2015	1.329	1.454	6.21	4.165	3.408	2.01	0.736	0.226	0.29
2016	1.342	1.549	6.6	4.52	3.722	2.139	0.776	0.254	0.312
2017	1.627	1.673	7.077	5.034	4.257	2.412	0.824	0.277	0.359
2018	1.792	1.813	7.485	5.608	4.818	2.689	0.91	0.309	0.394
2019	1.905	1.933	7.922	6.033	5.207	2.897	0.979	0.33	0.421
2020	2.003	1.938	8.176	6.094	5.447	2.922	1.009	0.338	0.444
2021	2.569	2.377	9.308	6.522	6.075	3.383	1.148	0.38	0.515

2.3.3 Calculation of grey correlation coefficient of panel data

The formula for calculating the grey relational coefficient of panel data is:

$$\gamma_i(s,t) = \frac{D + \rho \cdot L}{|d_i x_i(s,t) - d_1 x_1(s,t)| + \rho \cdot L} \quad (5)$$

where, $d_i x_i$ is the panel data of the reference after initialization, given the grey resolution; ρ is the distinguishing coefficient, and its value ranges from 0 to 1. In this study, $\rho = 0.5$. This value ensures sufficient discrimination of the grey relational coefficient while effectively reducing the impact of outliers. It is the most commonly used value in grey relational analysis. The above equation is the grey correlation coefficient of the initialized panel data $d_i x_i$ and $d_1 x_1$. In addition, $i = 1, 2, \dots, N$; $S = 1, 2, \dots, S$; and $t = 1, 2, \dots, T$.

$$D = \min_{i,s,t} |d_i x_i(s,t) - d_1 x_1(s,t)| \quad (6)$$

$$L = \max_{i,s,t} |d_i x_i(s,t) - d_1 x_1(s,t)| \quad (7)$$

Tables 3–7 show the computed grey correlation coefficient for the panel data following initialization.

Table 3. Grey correlation coefficient between regional gross domestic product and carbon emissions

Year	Shanxi	Inner Mongolia	Shandong	Henan	Sichuan	Shaanxi	Gansu	Qinghai	Ningxia
2010	1	0.953	0.632	0.724	0.729	0.859	0.96	0.985	0.989
2011	0.959	0.917	0.581	0.669	0.671	0.809	0.938	0.979	0.973
2012	0.959	0.924	0.559	0.637	0.651	0.786	0.935	0.981	0.979
2013	0.955	0.976	0.506	0.596	0.622	0.755	0.922	0.978	0.98
2014	0.946	0.993	0.483	0.576	0.594	0.733	0.91	0.974	0.985
2015	0.932	0.983	0.457	0.547	0.574	0.725	0.907	0.971	0.989
2016	0.936	0.96	0.436	0.518	0.544	0.702	0.894	0.967	0.997
2017	0.888	0.959	0.409	0.479	0.503	0.667	0.883	0.959	0.985
2018	0.868	0.962	0.397	0.443	0.463	0.634	0.87	0.95	0.986
2019	0.856	0.974	0.382	0.417	0.443	0.617	0.856	0.944	0.981
2020	0.845	0.999	0.371	0.416	0.429	0.617	0.855	0.94	0.978
2021	0.756	0.893	0.333	0.397	0.4	0.578	0.834	0.934	0.992

Table 4. Grey correlation coefficient of urban population and carbon emissions

Year	Shanxi	Inner Mongolia	Shandong	Henan	Sichuan	Shaanxi	Gansu	Qinghai	Ningxia
2010	1	0.922	0.775	0.794	0.737	0.882	0.936	0.98	0.989
2011	0.989	0.864	0.773	0.774	0.725	0.875	0.935	0.979	0.966
2012	0.973	0.849	0.794	0.763	0.739	0.882	0.942	0.985	0.967
2013	0.976	0.875	0.748	0.734	0.731	0.874	0.94	0.985	0.965
2014	0.99	0.873	0.742	0.737	0.719	0.872	0.936	0.984	0.968
2015	0.979	0.875	0.733	0.712	0.699	0.862	0.927	0.985	0.972
2016	0.98	0.876	0.712	0.694	0.681	0.847	0.917	0.986	0.977

2017	0.994	0.855	0.687	0.672	0.669	0.838	0.909	0.982	0.956
2018	0.991	0.826	0.7	0.656	0.651	0.83	0.91	0.979	0.95
2019	0.984	0.798	0.707	0.634	0.644	0.831	0.908	0.977	0.941
2020	0.979	0.781	0.693	0.627	0.634	0.831	0.91	0.974	0.934
2021	0.963	0.781	0.69	0.626	0.63	0.839	0.914	0.978	0.931

Table 5. Grey correlation coefficient between year-end resident population and carbon emissions

Year	Shanxi	Inner Mongolia	Shandong	Henan	Sichuan	Shaanxi	Gansu	Qinghai	Ningxia
2010	1	0.898	0.791	0.711	0.685	0.872	0.894	0.977	0.989
2011	0.98	0.837	0.801	0.71	0.685	0.872	0.897	0.978	0.964
2012	0.955	0.82	0.836	0.711	0.706	0.89	0.909	0.985	0.964
2013	0.95	0.84	0.8	0.698	0.71	0.892	0.913	0.986	0.96
2014	0.955	0.835	0.807	0.712	0.709	0.896	0.916	0.987	0.961
2015	0.977	0.834	0.819	0.704	0.7	0.895	0.914	0.988	0.963
2016	0.969	0.832	0.818	0.699	0.694	0.887	0.991	0.991	0.965
2017	0.948	0.809	0.803	0.691	0.692	0.886	0.909	0.989	0.943
2018	0.929	0.781	0.829	0.687	0.684	0.885	0.916	0.988	0.936
2019	0.916	0.754	0.843	0.676	0.689	0.895	0.917	0.988	0.925
2020	0.905	0.737	0.837	0.679	0.686	0.901	0.924	0.985	0.917
2021	0.889	0.736	0.845	0.684	0.688	0.917	0.932	0.99	0.913

Table 6. Grey correlation coefficient between added value of the secondary industry and carbon emissions

Year	Shanxi	Inner Mongolia	Shandong	Henan	Sichuan	Shaanxi	Gansu	Qinghai	Ningxia
2010	1	0.886	0.693	0.766	0.792	0.893	0.983	0.997	0.978
2011	0.949	0.85	0.651	0.714	0.739	0.839	0.967	0.993	0.962
2012	0.967	0.852	0.645	0.691	0.73	0.816	0.97	0.997	0.965
2013	0.98	0.886	0.598	0.658	0.702	0.791	0.965	0.996	0.963
2014	0.989	0.892	0.586	0.647	0.685	0.775	0.96	0.994	0.966
2015	0.975	0.898	0.572	0.625	0.677	0.787	0.973	0.994	0.968
2016	0.962	0.908	0.561	0.603	0.667	0.775	0.971	0.991	0.972
2017	0.982	0.894	0.534	0.564	0.642	0.739	0.968	0.984	0.96
2018	0.981	0.879	0.538	0.544	0.606	0.71	0.962	0.977	0.956
2019	0.975	0.861	0.535	0.523	0.589	0.701	0.958	0.974	0.948
2020	0.975	0.845	0.529	0.538	0.582	0.72	0.967	0.972	0.942
2021	0.827	0.972	0.474	0.52	0.542	0.66	0.945	0.966	0.959

Table 7. Grey correlation coefficient between energy structure and carbon emissions

Year	Shanxi	Inner Mongolia	Shandong	Henan	Sichuan	Shaanxi	Gansu	Qinghai	Ningxia
2010	1	0.875	0.871	0.978	0.915	0.864	0.839	0.888	0.762
2011	0.98	0.933	0.855	0.985	0.94	0.861	0.843	0.908	0.774
2012	0.955	0.959	0.818	1	0.975	0.877	0.858	0.904	0.78
2013	0.942	0.934	0.852	0.989	0.997	0.882	0.865	0.905	0.783
2014	0.946	0.939	0.844	0.98	0.987	0.888	0.868	0.912	0.78
2015	0.959	0.941	0.813	0.986	0.976	0.887	0.866	0.907	0.781
2016	0.952	0.951	0.808	0.985	0.974	0.873	0.868	0.87	0.783
2017	0.922	0.981	0.806	0.982	0.964	0.875	0.876	0.887	0.793
2018	0.896	0.971	0.776	0.982	0.968	0.879	0.886	0.907	0.794
2019	0.879	0.935	0.758	0.991	0.952	0.893	0.894	0.916	0.805
2020	0.868	0.903	0.752	0.983	0.953	0.891	0.899	0.922	0.81
2021	0.852	0.899	0.74	0.959	0.945	0.912	0.897	0.908	0.816

2.3.4 Calculation of grey correlation degree of panel data

Definition 2: Let the grey correlation coefficient between the panel data x_i and the reference panel x_i be denoted by, with Eq. (8) representing the grey correlation degree of the panel data, with $i = 1, 2, \dots, N$.

Definition 3:

$$r_{i,t} = \frac{1}{S} \sum_{s=1}^S \sum_{T=1}^T \gamma_i(s, t) \quad (8)$$

$$r_{i,t} = \frac{1}{S} \sum_{s=1}^S \gamma_i(s, t) \quad (9)$$

where, $i = 1, 2, \dots, N$; and $t = 1, 2, \dots, T$. The time-series grey correlation coefficient for the panel data is represented by $r_{i,t}$, which measures the average grey correlation coefficients among different subjects at the corresponding time points.

Definition 4:

$$r_{i,s} = \frac{1}{T} \sum_{t=1}^T \gamma_i(s, t) \quad (10)$$

where, $i = 1, 2, \dots, N$; and $t = 1, 2, \dots, T$. The cross-sectional grey correlation coefficient of the panel data quantifies the average grey correlation coefficients related to the same topic across different time points.

Figure 1, Table 8, and Table 9 show the time-series grey correlation degrees of the panel data.

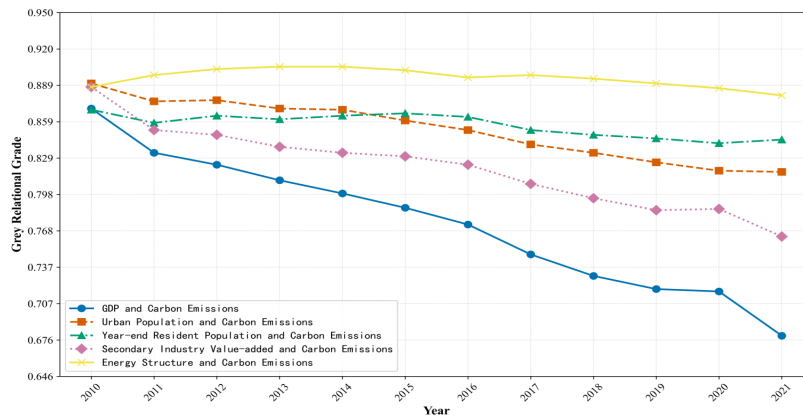


Figure 1. Time-series grey correlation degrees between influencing factors and carbon emissions (2010–2021)

Table 8. Cross-sectional grey correlation degree of the panel data

Region	Provincial Gross Domestic Product and Carbon Emissions	Urban Population and Carbon Emissions	Resident Population and Carbon Emissions at the End of the Year	Added Value and Carbon Emissions of the Secondary Industry	Energy Mix and Carbon Emissions
Shanxi	0.908	0.983	0.948	0.964	0.929
Inner Mongolia	0.958	0.848	0.809	0.885	0.935
Shandong	0.462	0.73	0.819	0.576	0.808
Henan	0.535	0.702	0.697	0.616	0.983
Sichuan	0.552	0.688	0.694	0.663	0.962
Shaanxi	0.707	0.855	0.891	0.767	0.882
Gansu	0.897	0.924	0.913	0.966	0.872
Qinghai	0.964	0.981	0.986	0.986	0.903
Ningxia	0.985	0.96	0.95	0.95	0.788

Table 9. Grey correlation degree of the panel data

Provincial Gross Domestic Product and Carbon Emissions	Urban Population and Carbon Emissions	Resident Population and Carbon Emissions at the End of the Year	Added Value and Carbon Emissions of the Secondary Industry	Energy Mix and Carbon Emissions
0.774	0.852	0.856	0.821	0.896

2.4 Explanation of Applicability and Limitations

The panel data grey relational analysis method is suitable for this study for three main reasons. First, early

statistical data are missing for some provinces in the Yellow River basin, and this method has low requirements on sample size. Second, it can simultaneously capture the temporal evolution and spatial variation of factors influencing carbon emissions. Third, it does not assume linear relationships between variables, which better fits the nonlinear nature of the carbon emission system.

At the same time, this method has certain limitations. It only analyzes the degree of correlation between factors and cannot reveal causal relationships. In addition, the value of the distinguishing coefficient is somewhat subjective. Future research could combine methods such as structural equation modeling and panel regression to further verify the causal relationships and transmission mechanisms between factors.

3. Case Analysis

3.1 Grey Correlation Coefficient of Panel Data

Six panel datasets (carbon emissions and five influencing factors: regional gross domestic product, urban population, year-end permanent resident population, added value of the secondary industry, and energy mix) were preprocessed using the initial value method for dimensionless treatment. Table 1 shows the dimensionless carbon emission data following this preprocessing, and the initialized regional gross domestic product data are shown in Table 2. It should be noted that a comprehensive review of the China Energy Statistical Yearbook for all provinces in the Yellow River basin confirmed that coal dominates regional energy consumption. Therefore, the energy mix indicator was explicitly defined as the proportion of coal consumption in total primary energy consumption.

Subsequently, the absolute differences between the panel matrix of each influencing factor and the reference carbon emission matrix were calculated to determine the grey resolution coefficients. Finally, the grey correlation coefficients between the five factor matrices and the reference matrix were obtained. The results are listed in Tables 3 to 7.

3.2 Grey Correlation Degree of Panel Data in Time Series

Based on the measured time-series grey relational degree data from 2010 to 2021, the time-series relational degrees between the five influencing factors and carbon emissions were calculated. The dynamic evolution characteristics are shown in Figure 1 and analyzed below.

From the perspective of temporal evolution, the grey relational degrees of all five factors with carbon emissions remained above 0.68, all falling into the strong correlation category. This confirms that the selected indicators are all core driving factors of carbon emissions in the Yellow River basin. The overall relational intensity showed a weakening trend year by year, indicating that the low-carbon transition policies in the basin have achieved phased results.

(i) The grey relational degree between regional gross domestic product and carbon emissions continuously decreased from 0.87 in 2010 to 0.68 in 2021, a cumulative decline of 21.8%, the largest drop among all factors. This indicates a significant decoupling trend between economic growth and carbon emissions in the basin. With the advancement of ecological civilization, industrial structure optimization, and energy efficiency improvement, the dependence of economic development on high-carbon energy consumption has greatly diminished.

(ii) The grey relational degree between energy structure and carbon emissions remained above 0.88 over the long term, only slightly fluctuating from 0.89 in 2010 to 0.88 in 2021. It was consistently the factor with the highest relational intensity. This confirms that the coal-dominated energy structure remains the primary driving factor of carbon emissions in the basin. The clean energy transition has progressed slowly, making the energy structure a critical bottleneck for carbon emission reduction.

(iii) The grey relational degree between year-end permanent resident population and carbon emissions gently decreased from 0.87 in 2010 to 0.84 in 2021, with small fluctuations and an overall stable trend. The slowing population growth and the weakening effect of natural increase in the basin have gradually reduced the direct pulling effect of population size on energy consumption and carbon emissions.

(iv) The grey relational degree between urban population and carbon emissions steadily decreased from 0.89 in 2010 to 0.81 in 2021, a cumulative decline of 8.3%. This reflects the remarkable effectiveness of low-carbon urbanization in the basin. The widespread adoption of green buildings, public transportation, and low-carbon lifestyles has weakened the synchronicity between urban population agglomeration and the growth of carbon emissions.

(v) The grey relational degree between the added value of the secondary industry and carbon emissions dropped significantly from 0.89 in 2010 to 0.76 in 2021, a cumulative decline of 14.1%. This indicates that the green transformation of industry has achieved outstanding results. The elimination of outdated capacity in energy-intensive industries and the application of energy-saving technologies have effectively reduced the carbon emission-driving effect of industrial development.

Overall, the relational intensity of most driving factors showed a declining trend year by year, confirming the

effectiveness of the low-carbon development path in the Yellow River basin. Among them, the energy structure still plays a dominant role, while the correlations between economic growth, urbanization, industrial development and carbon emissions have significantly weakened. This indicates that the basin is steadily transitioning from high-carbon development to low-carbon development.

3.3 Cross-Sectional Grey Correlation Degree of Panel Data

The cross-sectional grey correlation degrees for all nine provinces are presented in Table 8. Based on the regional heterogeneity of correlation intensity, the carbon emission driving patterns in the Yellow River basin can be categorized into three distinct types.

(i) Energy-industry-dominated type. This type includes Shanxi, Inner Mongolia, Gansu, Qinghai, and Ningxia. In these provinces, the grey relational degrees of both energy structure and the added value of the secondary industry exceeded 0.85, making them the core driving factors of carbon emissions. The reason is that these provinces serve as major national energy bases. Their industrial structure is dominated by energy-intensive industries such as coal mining, coal chemical processing, and non-ferrous metal smelting, and their economic development relies heavily on fossil fuel consumption.

(ii) Population-urbanization-dominated type. This type includes Shandong and Shaanxi. In these two provinces, the grey relational degrees of urban population and year-end permanent resident population were higher than those of other factors. Shandong is a populous and economically strong province with an urbanization rate exceeding 65%. Urban infrastructure construction and residential consumption have become the main sources of carbon emission growth. Shaanxi is currently in a stage of rapid urbanization, and population agglomeration in cities has driven a rapid increase in energy consumption.

(iii) Multi-factor integrated type. This type includes Henan and Sichuan. In these two provinces, the grey relational degrees of energy structure, population, and industrial structure are relatively balanced. Henan is a major agricultural and populous province, where industries are mainly food processing and equipment manufacturing, and carbon emissions are jointly influenced by multiple factors. Sichuan has abundant hydropower resources and a relatively optimized energy structure, but its large population size and rapid urbanization process make the driving factors of carbon emissions more complex.

3.4 Grey Correlation Degree of Panel Data

The comprehensive relational degrees between each factor and carbon emissions were calculated using Eq. (8), and the results are presented in Table 9. Overall, the driving factors of carbon emissions in the Yellow River basin are ranked as follows: energy structure > year-end permanent population > urban population > added value of the secondary industry > regional gross domestic product. This finding suggests that energy structure is the primary driver of carbon emissions in the basin. “Coal control” remains the core task for future carbon emission reduction. Population size and urbanization have a significant impact on carbon emissions. Population urbanization and low-carbon city construction need to be promoted in a coordinated way. The secondary industry still exerts a considerable influence on carbon emissions. The green transformation of industry must be continuously advanced. The correlation between economic growth and carbon emissions has become the lowest. The basin’s economic development is gradually reducing its dependence on high carbon emissions.

3.5 Analysis of Interactions among Variables

Existing analyses often treat influencing factors independently. In reality, complex interactions exist among these factors.

(i) Interaction between energy structure and industrial structure: The development of energy-intensive industries directly drives the growth of coal consumption. Conversely, the clean transition of the energy structure can also push forward industrial structure upgrading. For example, in Shanxi Province, a 1% decrease in the share of coal consumption can reduce the carbon emission intensity of the secondary industry by about 0.8%.

(ii) Interaction between population and energy structure: Population growth not only directly increases residential energy consumption, but also indirectly raises fossil energy demand by stimulating industrial production. In regions where coal dominates the energy structure, the amplifying effect of population growth on carbon emissions is more pronounced.

(iii) Interaction between economic development and technological progress: Economic growth provides financial support for technology research and development. Technological progress, in turn, improves energy efficiency and lowers the carbon emission intensity of economic growth. In the Yellow River basin, economically developed provinces such as Shandong and Henan show significantly higher energy efficiency than less developed western provinces.

4. Policy Recommendations

Based on the above research findings and in line with the strategic requirements for ecological protection and high-quality development in the Yellow River basin, the following differentiated carbon emission reduction policy recommendations are proposed.

(i) The clean transition of energy structure by category should be promoted. Energy structure is the primary driver of carbon emissions in the Yellow River basin. Differentiated transition strategies should be implemented based on the resource endowments of each province. As for energy-rich areas (Shanxi, Inner Mongolia, and Ningxia), total coal consumption should be strictly controlled and the clean and efficient use of coal should be accelerated. Focusing on developing the modern coal chemical industry, the industrial chain should be extended, and product added value should be increased. Meanwhile, the advantages of wind and solar energy resources should be made full use of. Large-scale wind power and photovoltaic bases should be built, and the integrated development of wind, solar, thermal power, and energy storage should be promoted. As for areas rich in hydropower resources (Sichuan, Qinghai, and Gansu), hydropower resources should be vigorously developed and the share of hydropower in energy consumption should be raised. At the same time, renewable energy sources such as solar and wind power should be developed, and a clean energy system with multi-energy complementarity should be built. As for the eastern coastal area (Shandong), the development of offshore wind power and nuclear power should be prioritized. The construction of coastal nuclear power bases should be advanced. The proportion of coal consumption should be gradually reduced, and a national demonstration province should be built for clean energy.

(ii) The green and low-carbon transformation of industry should be promoted. In view of the significant impact of the secondary industry on carbon emissions, the optimization and upgrading of the industrial structure should be accelerated. As for transformation of traditional energy-intensive industries, energy-saving and carbon-reduction retrofits in industries such as steel, chemicals, and non-ferrous metals should be implemented. The application of advanced energy-saving technologies and equipment should be promoted to improve energy efficiency. Enterprises should be supported in piloting circular economy practices to achieve efficient resource recycling. As for the cultivation of emerging low-carbon industries, strategic emerging industries such as new energy, new materials, and high-end equipment manufacturing should be vigorously developed, and new economic growth points should be cultivated. Provinces such as Shandong and Henan should be supported in building national-level green manufacturing bases. The transformation of industries towards high-end, intelligent, and green development should be promoted. As for strict industrial access management, the approval of projects with high energy consumption and high emissions should be strictly controlled for provinces at the early stage of industrialization, such as Inner Mongolia, Gansu, and Ningxia. Following the old path of “pollution first, treatment later” should be avoided.

(iii) Population urbanization and low-carbon development should be coordinated. In view of the impact of population and urbanization on carbon emissions, green urbanization should be promoted. As for the urban spatial layout optimization, a compact urban spatial structure should be built to reduce commuting distances. Green travel modes such as public transport, cycling, and walking should be promoted to lower transportation carbon emissions. As for the promotion of green buildings, energy-saving standards for new buildings should be raised and energy-saving retrofits of existing buildings should be advanced. The use of green building materials and renewable energy application technologies in buildings should be encouraged to reduce operational energy consumption. As for green consumption guidance, publicity and education should be strengthened to raise residents’ low-carbon awareness. Residents should be encouraged to purchase energy-saving home appliances and new energy vehicles. A simple, moderate, green, and low-carbon lifestyle should be advocated.

(iv) A collaborative emission reduction mechanism should be established for the basin. Carbon emissions in the Yellow River basin show significant regional interconnectedness. A cross-regional collaborative emission reduction mechanism needs to be established. First, a carbon emission trading market for the basin should be established. The Yellow River basin should be integrated into the national carbon emission trading system. The establishment of a cross-regional trading mechanism for carbon emission rights within the basin should be explored so as to minimize emission reduction costs. Second, regional science and technology cooperation should be strengthened. A basin-level low-carbon technology innovation alliance should be established. Key technologies such as carbon capture, utilization, and storage should be jointly tackled. The application and dissemination of advanced low-carbon technologies within the basin should be promoted. Finally, the ecological compensation mechanism should be improved. An ecological compensation system between the upstream and downstream areas of the basin should be established. Financial and policy support should be provided for regions with outstanding contributions to ecological protection and carbon emission reduction. Common development of the basin should be achieved.

Author Contributions

Formal analysis, Y.L.; resources, B.J.Z.; data curation, A.X.L.; writing—original draft preparation, Y.J.H.; writing—review and editing, H.H.; visualization, B.W.Z.; supervision, W.X.D.; project administration, Y.W.H. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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